1. Data Flow Diagram (DFD)

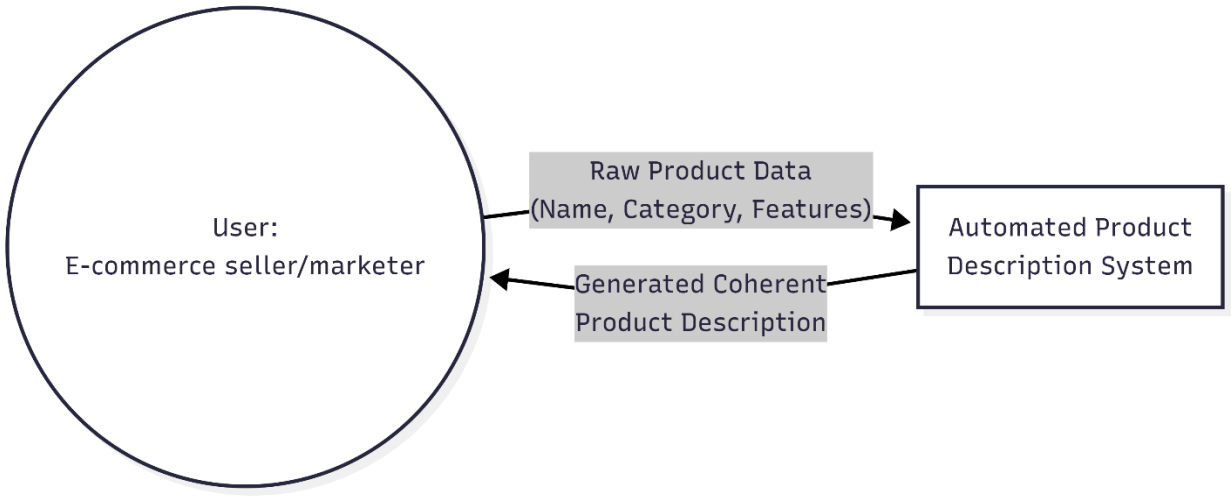
This shows how data moves through your system, from the user input to the final output.

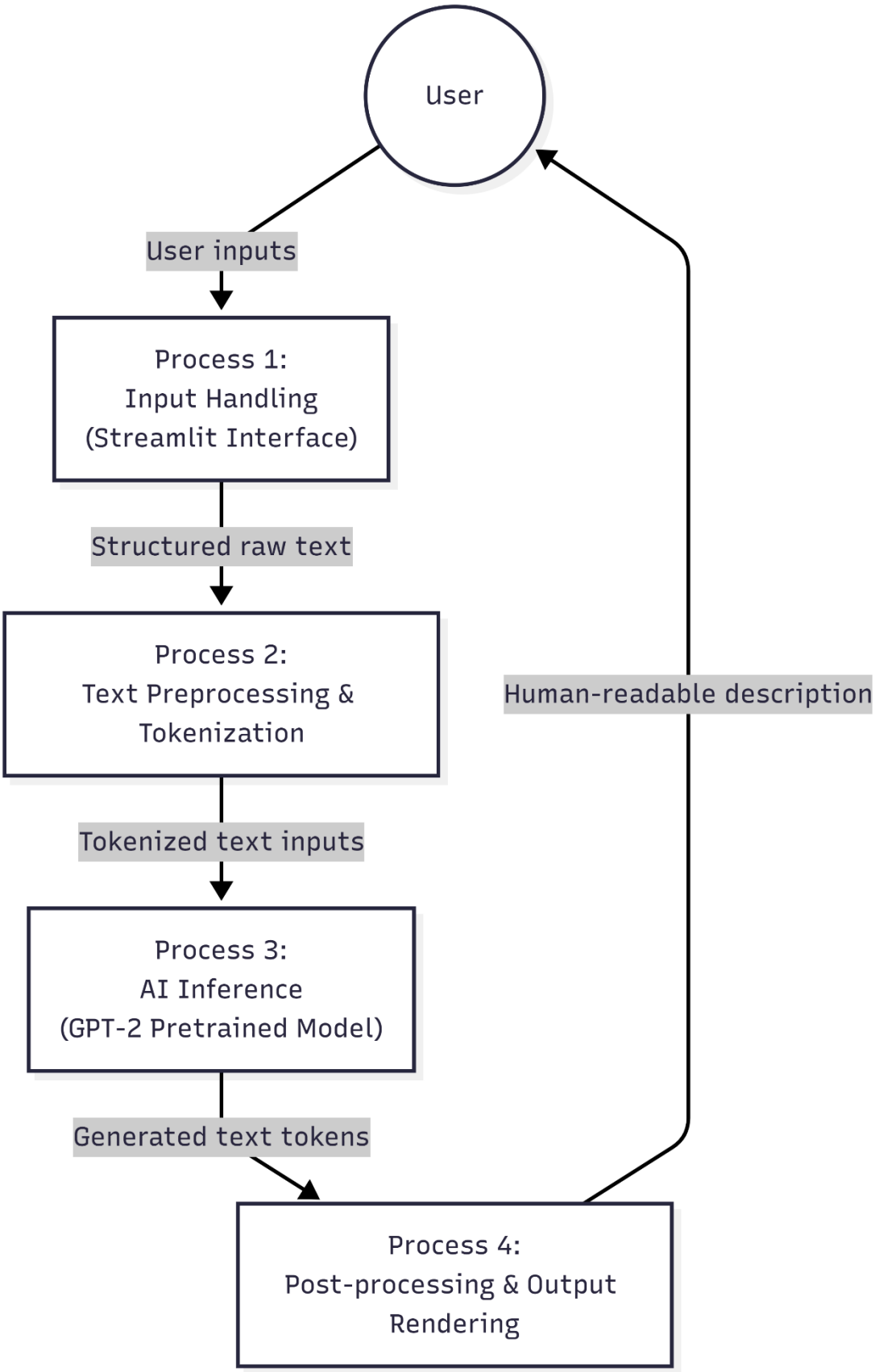
Context Level (Level 0):

- **External Entity:** User (E-commerce seller/marketer).
- **Main Process:** Automated Product Description System.
- **Data Flow In:** Raw Product Data (Name, Category, Features).
- **Data Flow Out:** Generated Coherent Product Description.

Level 1 (Detailed System Flow):

- **Process 1: Input Handling (Streamlit Interface).**
  - *In:* User inputs (Name, Features).
  - *Out:* Structured raw text.
- **Process 2: Text Preprocessing & Tokenization.**
  - *In:* Structured raw text.
  - *Out:* Tokenized text inputs (Numbers).
- **Process 3: AI Inference (GPT-2 Pretrained Model).**
  - *In:* Tokenized text.
  - *Out:* Generated text tokens.
- **Process 4: Post-processing & Output Rendering.**
  - *In:* Generated text tokens.
  - *Out:* Human-readable Product Description displayed to the user.

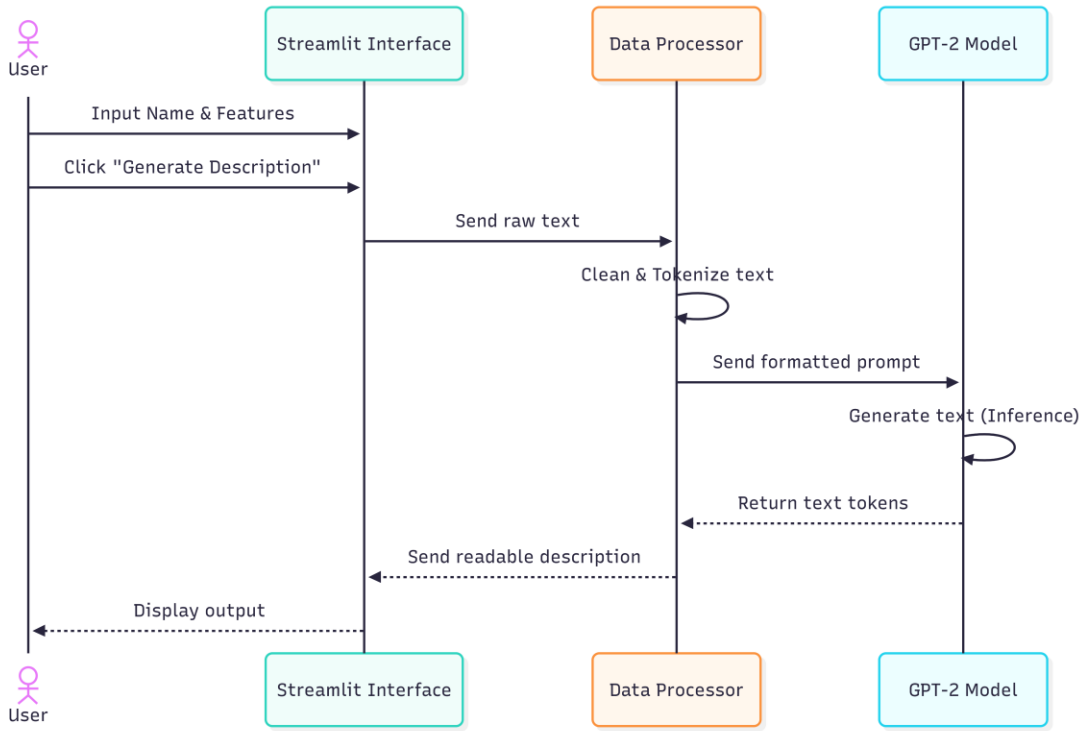



User
User inputs
Process 1:
Input Handling
(Streamlit Interface)
Structured raw text
Process 2:
Text Preprocessing &
Tokenization
Tokenized text inputs
Process 3:
AI Inference
(GPT-2 Pretrained Model)
Generated text tokens
Process 4:
Post-processing & Output
Rendering
Human-readable description

2. Sequence Diagram

This represents the process flow and key interactions between your components over time.

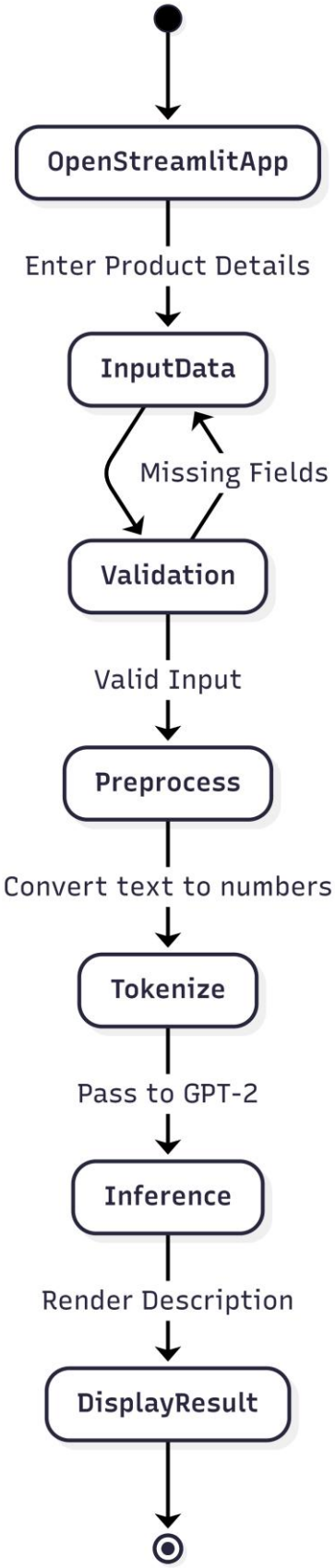
- **Actors/Lifelines:** User, Streamlit App , Backend Controller, GPT-2 Model.
- **Sequence of Events:**
    1. **User** -> **Streamlit App**: Enters product name and key features.
    2. **User** -> **Streamlit App**: Clicks "Generate Description".
    3. **Streamlit App** -> **Backend Controller**: Sends input data payload.
    4. **Backend Controller** -> **Backend Controller**: Formats the prompt using predefined templates.
    5. **Backend Controller** -> **GPT-2 Model**: Sends formatted prompt for inference.
    6. **GPT-2 Model** ->**GPT-2 Model**: Processes tokens and generates description.
    7. **GPT-2 Model** -> **Backend Controller**: Returns generated text output.
    8. **Backend Controller** -> **Streamlit App**: Sends final formatted text.
    9. **Streamlit App** -> **User**: Displays the marketing-ready description on the screen.



3. Activity Diagram

This visualizes the workflow of user actions and system processes.

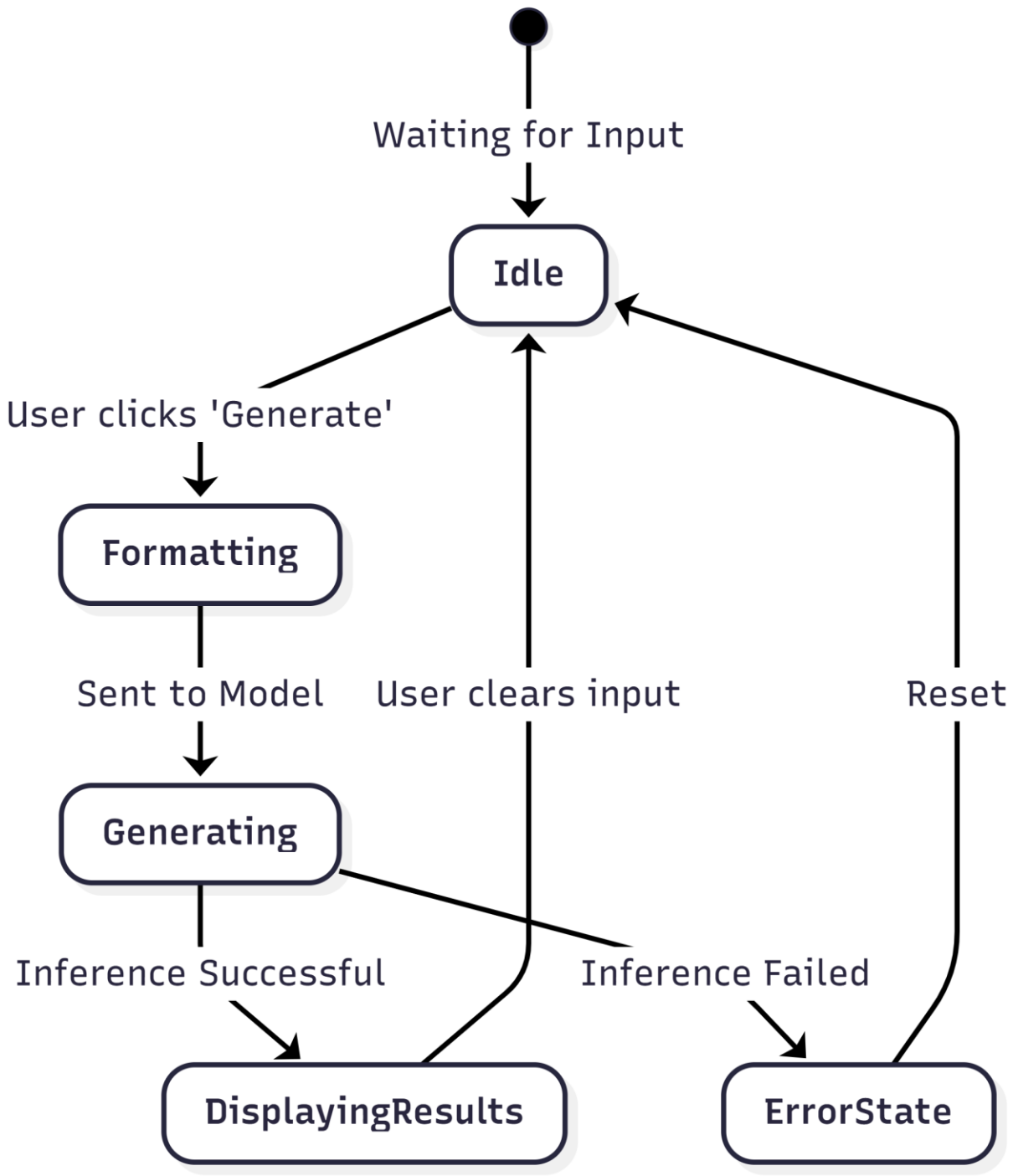
- **Start Node** (Solid black circle)
- **Action:** User opens the Streamlit Web Application.
- **Action:** User inputs Product Name, Category, and Features.
- **Decision Node (Diamond):** Are all required fields filled?
    - *If No:* **Action:** Display validation error -> Loop back to input stage.
    - *If Yes:* **Action:** System formats the input into a cohesive prompt.
- **Action:** Tokenize text (convert to numbers).
- **Action:** Pass tokenized data to the fine-tuned GPT-2 Model.
- **Action:** Model generates the product description.
- **Action:** Display the final description on the Streamlit UI.
- **End Node** (Target circle)


OpenStreamlitApp
Enter Product Details
InputData
Missing Fields
Validation
Valid Input
Preprocess
Convert text to numbers
Tokenize
Pass to GPT-2
Inference
Render Description
DisplayResult

4. State Diagram

This represents the different states of your Streamlit application during the user session.

- **Initial State:** Idle (App is loaded, waiting for user input).
- **Transition:** User inputs data and clicks 'Generate'.
- **State:** Processing/Formatting (System structures the data into a prompt).
- **Transition:** Prompt is sent to the model.
- **State:** Generating_Description (Model is performing inference).
- **Transition:** Model completes generation successfully.
    - *(Alternative Transition):* Model fails or times out -> **State:** Error -> Return to Idle.
- **State:** Displaying_Results (The coherent product description is shown on screen).
- **Transition:** User clicks 'Reset' or 'New Product'.
- **Return to Initial State:** Idle.


Waiting for Input
Idle
User clicks 'Generate'
Formatting
Sent to Model
User clears input
Reset
Generating
Inference Successful
Inference Failed
DisplayingResults
ErrorState

5. Class Diagram

This defines the structure of your system's software components, attributes, and methods.

Class: StreamlitUI

- ***Attributes:***
    - user_inputs: Dictionary
    - is_loading: Boolean
- ***Methods:***
    - render_input_form()
    - capture_data()
    - display_output(description: String)

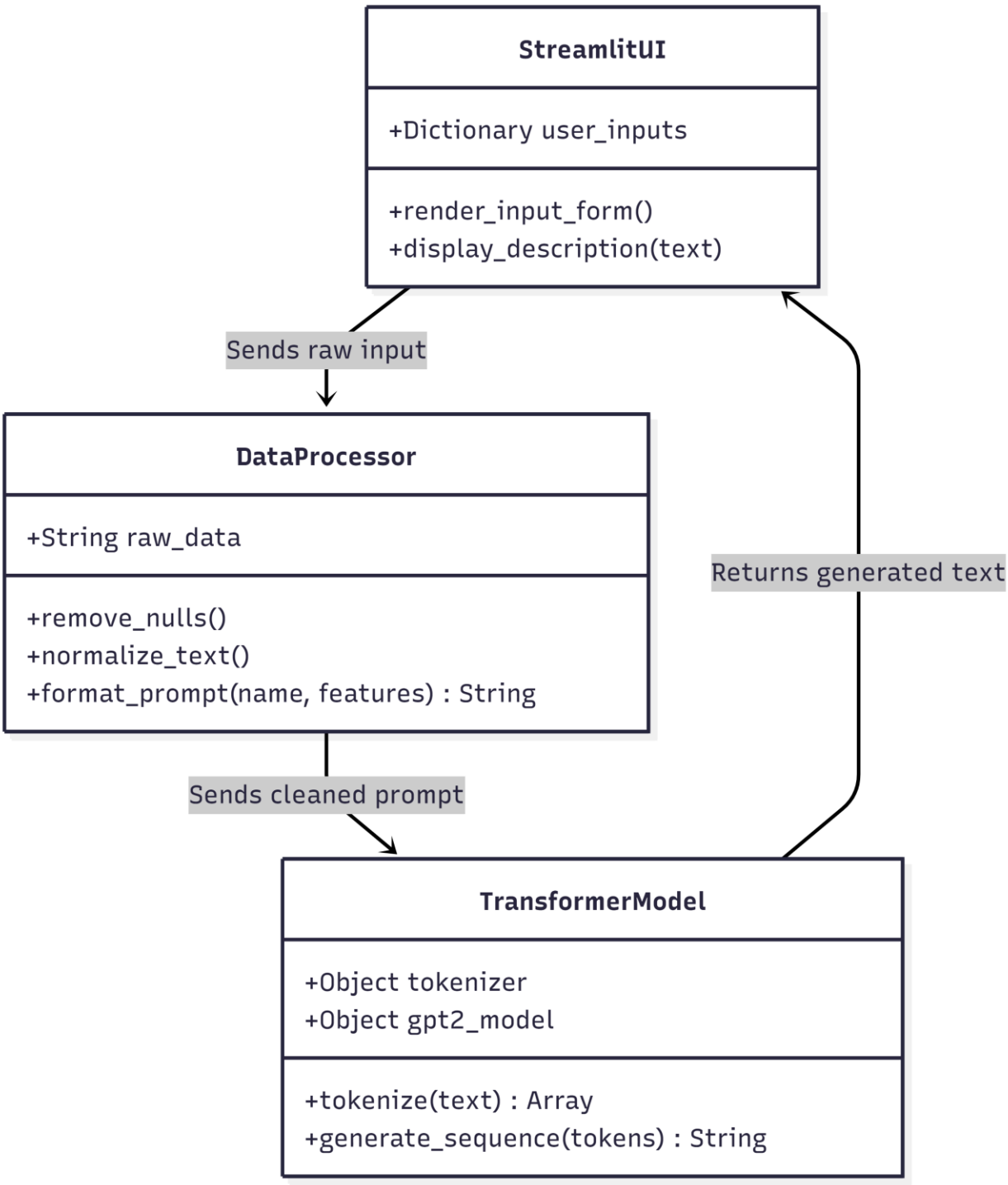
Class: DataProcessor

- ***Attributes:***
    - raw_text: String
- ***Methods:***
    - clean_text(text: String): String (Removes nulls, normalizes)
    - format_prompt(name: String, features: String): String

Class: TransformerModel

- ***Attributes:***
    - model_weights: File (Saved Model)
    - tokenizer: Object
- ***Methods:***
    - load_model()
    - tokenize_input(prompt: String): Array
    - generate_text(tokens: Array): String

Relationships:

- StreamlitUI depends on DataProcessor to clean and format the input.

- StreamlitUI calls TransformerModel to fetch the generated description.